Envelope bounds for the face transfer constants

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Abstract

Astra #9 R5, step 3 (face half). The explicit face constant of unit 131 involves tail moments of the actual face coefficients. Under the envelope hypotheses of unit 110 (Taylor data f_m and remainder envelope H bounded by $C(1+s)^D e^{\beta s a'}$), every face coefficient is bounded pointwise by an explicit multiple of the Gaussian envelope $(1+s)^D e^{\beta s a'}$, tail moments are monotone under pointwise envelopes, and hence faceConst $\leq$ faceEnvConst $\cdot \int_0^\infty s^{q-1}(1+|\log s|)e^{-\beta s^2}(1+s)^D e^{\beta s a'} ds$ with faceEnvConst a function of $(p_2, b, h_1, k_1, k_2, M, C_f, C_H)$ only (`faceConst_le_env`). This is the ingredient that makes the cutoff constant uniform over a family of amplitudes with a common envelope. Zero `sorry`/`axiom`.

1 The things formalised

```
noncomputable def gaussEnv (  a' :  ) (D :  ) (s :  ) :   :=
  (1 + s) ^ D * Real.exp (  * s * a')
theorem tailMoment_le_of_env (  a' q K :  ) (j D :  ) (c :   →  ) (h  : 0 <  ) (hq : 0 < q)
    (hc : Measurable c) (henv :   s, 0 < s → |c s|   K * gaussEnv   a' D s) :
    tailMoment   q j c   K * envMoment   q j (gaussEnv   a' D)
theorem envMoment_gauss_mono (  a' q :  ) (D :  ) (h  : 0 <  ) (hq : 0 < q) :
    envMoment   q 0 (gaussEnv   a' D)   envMoment   q 1 (gaussEnv   a' D)
noncomputable def fpEnvConst (  b :  ) (k  M :  ) (Cf CH :  ) :   :=
  1 / (k  :  ) * (CH / ((M :  ) -  ) * b ^ ((M :  ) -  )
    +   m   Finset.range M, Cf * |axisPrim   b m|)
theorem faceFPCoeff_abs_le_env ... (s :  ) (hs : 0 < s) :
    |faceFPCoeff   b k  Ψ f M s|   fpEnvConst   b k  M Cf CH * gaussEnv   a' D s
noncomputable def faceCEnv (  b :  ) (k  k  M :  ) (Cf CH :  ) : Fin M   Fin 2 →   :=
  Sum.elim (fun m => |faceScalar   (b ^ k ) k  k  m| * Cf)
    ![fpEnvConst   b k  M Cf CH, (M :  ) * (1 / ((k  :  ) * k )) * Cf]
theorem faceC_abs_le_env ... (i : Fin M   Fin 2) (s :  ) (hs : 0 < s) :
    |faceC   b k  k  Ψ f M i s|   faceCEnv   b k  k  M Cf CH i * gaussEnv   a' D s
noncomputable def faceEnvConst (p  b :  ) (h  k  k  M :  ) (Cf CH :  ) :   :=
  envEnvConst   (b ^ k ) k  k  M CH
    +   i, (b ^ (k  + k )) ^ (face  p   k  M i - q) * faceCEnv   b k  k  M Cf CH i
theorem faceConst_le_env (  p  b a' Cf CH :  ) (h  k  k  M D :  ) ...
    (hCf : 0   Cf) (hCH : 0   CH)
    (hfenv :   m, m < M →   s, 0 < s → |f m s|   Cf * gaussEnv   a' D s)
    (hHenv :   s, 0 < s → H s   CH * gaussEnv   a' D s) (hrem : ...) :
    faceConst   p  b h  k  k  M Ψ f H
       faceEnvConst p  b h  k  k  M Cf CH
       * envMoment   (p  + ((M :  ) - ((k  :  ) * p  - h  - 1)) / k ) 1 (gaussEnv   a' D)
```

2 Mathematics

The monomial coefficient is faceScalar $\cdot f_m$, the finite-part coefficient is bounded via $|\mathrm{FP}_\gamma| \le \frac{H}{M-\gamma} b^{M-\gamma} + \sum_m |f_m||\mathrm{axisPrim}(m)|$ (unit 110), the log coefficient by $\frac{M}{k_1 k_2} \max |f_m|$, and the remainder envelope faceEnv is a constant multiple of H; each therefore has the envelope $K\,(1+s)^D e^{\beta s a'}$ with an explicit K. A tail moment $\int s^{q-1}(1+|\log s|)^j e^{-\beta s^2}|c|$ of such a c is at most K times the Gaussian envelope moment (pointwise comparison of integrable integrands), and the $j=0$ moment is at most the $j=1$ moment. Summing with the (nonnegative) weights $(b^{k_1+k_2})^{\alpha_i - q}$ gives the bound.

3 Paper-facing meaning

Combined with units 130–131, the $d=2$ cutoff remainder constant is now bounded by an explicit function of the envelope data $(C_0, L, D, \rho, b, \beta, h, k, M)$ times Gaussian moments, once the face envelopes of the analytic adapter ($C_f = C_0\rho^{-i}(1+\rho^{-1})^{M_2}$, $C_H = C_0\rho^{-i-M_2}/(1-b/\rho)$, corners $C_f = C_0\rho^{-i-j}$, $C_H = 0$) are substituted. That substitution and the resulting uniform-family theorem ($\exists K\ \forall c$ with the common envelope, $\forall N \ge 1$) is the next unit; it delivers Astra #9's R5 target and the input for the parameter integration over the noncritical coordinates.

4 Lean notes

- Exporting the local bounds of a long proof as standalone lemmas is cheap and makes them reusable; the envelope function was given a name (`gaussEnv`) so that all bounds share one shape.
- `moment_integrableOn_of_envelope` expects the bound as `K * (1 + s) ^ D * exp`, left-associated; a hypothesis stated with `K * gaussEnv` needs `rw [mul_assoc]` in the discharging lambda.
- Entries of `![a, b]` at `Fin.mk` indices produced by `fin_cases` are best handled by `show` with the expected entry (definitional), not by `Matrix.cons_val_zero/one` in `simp only`.
- `positivity` cannot see through `(M :  ) -  `; supply `div_nonneg` and `mul_nonneg` explicitly.
- `gcongr` for a power comparison leaves `1   1 + |log s|` and sometimes `0   1`; `all_goals linarith` covers both.